C10 Group 7, the Halogens – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Properties and Trends **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Appearance of F_2 at room temperature				Pale yellow gas
2	Appearance of Cl_2 at room temperature				Pale green gas
3	Appearance of Br_2 at room temperature				Red-brown liquid
4	Appearance of I_2 at room temperature				Grey-black solid
5	Colour of iodine vapour				Purple
6	Trend in boiling point down Group 7				Increases
7	Trend in electronegativity down Group 7				Decreases
8	Trend in oxidising ability of halogens down Group 7				Decreases
9	Trend in reducing ability of halide ions down Group 7				Increases
10	Why boiling point increases down Group 7				More electrons, stronger van der Waals

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C10 Group 7, the Halogens – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Displacement Reactions **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Strongest oxidising halogen				F_2
2	Strongest reducing halide ion				I^-
3	Colour of $Cl_2(aq)$				Pale green
4	Colour of $Br_2(aq)$				Orange
5	Colour of $I_2(aq)$				Brown
6	Colour of Br_2 in hexane				Orange
7	Colour of I_2 in hexane				Purple
8	Ionic equation for Cl_2 with Br^-				$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$
9	Ionic equation for Cl_2 with I^-				$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$
10	Ionic equation for Br_2 with I^-				$Br_2 + 2I^- \rightarrow 2Br^- + I_2$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Silver Nitrate Test for Halides **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Reagent to test for halide ions				Acidified $AgNO_3$
2	Acid used to acidify				Dilute HNO_3
3	Why the $AgNO_3$ is acidified				Removes CO_3^{2-} so Ag_2CO_3 cannot form
4	Why HCl is not used to acidify				Adds Cl^- ions
5	Colour of AgCl precipitate				White
6	Colour of AgBr precipitate				Cream
7	Colour of AgI precipitate				Yellow
8	Result with F^- ions				No precipitate
9	Silver halide that dissolves in dilute NH_3				AgCl
10	Silver halide insoluble even in conc NH_3				AgI

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 4: Halides with Concentrated Sulfuric Acid **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Gas formed from NaCl with conc H_2SO_4				HCl
2	Observation for NaCl with conc H_2SO_4				Steamy fumes
3	Gases formed from NaBr with conc H_2SO_4				HBr, Br_2, SO_2
4	Observation for NaBr with conc H_2SO_4				Steamy fumes and orange-brown fumes
5	Reduction product of H_2SO_4 with Br^-				SO_2
6	Reduction products of H_2SO_4 with I^-				SO_2, S, H_2S
7	Observation showing S forms				Yellow solid
8	Smell of H_2S				Rotten eggs
9	Observation showing I_2 forms				Purple fumes, black solid
10	Why Cl^- does not reduce H_2SO_4				Cl^- is too weak a reducing agent

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C10 Group 7, the Halogens – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 5: Uses of Chlorine **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Equation for Cl_2 with water				$Cl_2 + H_2O \rightleftharpoons HCl + HClO$
2	Name of HClO				Chloric(I) acid
3	Type of reaction of Cl_2 with water				Disproportionation
4	Equation for Cl_2 with water in sunlight				$2Cl_2 + 2H_2O \rightarrow 4HCl + O_2$
5	Why chlorine is added to drinking water				Kills bacteria
6	Risk of chlorinating water				Forms toxic chlorinated hydrocarbons
7	Why chlorine is still used in water				Benefits outweigh risks
8	Equation for Cl_2 with cold dilute NaOH				$Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$
9	Name of NaClO				Sodium chlorate(I)
10	Use of NaClO solution				Bleach

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C10 Group 7, the Halogens – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 6: Equations **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Equation for NaCl with conc H_2SO_4				$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
2	Equation for NaBr with conc H_2SO_4				$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$
3	Equation for HBr reducing H_2SO_4				$2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$
4	Equation for HI reducing H_2SO_4 to SO_2				$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$
5	Equation for HI reducing H_2SO_4 to S				$6HI + H_2SO_4 \rightarrow 3I_2 + S + 4H_2O$
6	Equation for HI reducing H_2SO_4 to H_2S				$8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$
7	Half-equation for oxidation of Br^-				$2Br^- \rightarrow Br_2 + 2e^-$
8	Ionic equation for the silver chloride precipitate				$Ag^+ + Cl^- \rightarrow AgCl$
9	Equation for AgCl dissolving in NH_3				$AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]^+ + Cl^-$
10	Ionic equation for Cl_2 with cold dilute NaOH				$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.